Physics-Based Modelling of Faults and System Efficiency Degradations in District Cooling Networks

October 1, 2026

1 Work Completed Since the Previous Report

The work since the previous progress report has focused on the physical modelling and validation of the five faults and system efficiency degradations (FSEDs) prioritised within the SWiM project: *poor balancing*, *too safe safety margins in control*, *clogging of water filters*, *scaling/fouling of heat exchangers*, and *stuck/sticking/leaking hydronic valves* [10]. The existing `dcngen`/DCN-Bench framework developed by Poulsen [9] was used as the simulation basis. The framework already contains the hydraulic and thermal network model and a representation of heat exchanger fouling. The present work implemented or extended representations for poor balancing, too safe safety margins in control, clogging of water filters, and the stuck case of the hydronic valve FSED. Heat exchanger fouling was retained as an inherited reference mechanism.

Before generating a larger synthetic dataset, each mechanism was tested individually. For each test, a faulted simulation was compared with a matched healthy simulation using the same network, load realisation and random seed. This provides a controlled comparison in which the intended FSED intervention is the main difference between the two simulations. The purpose of these tests is to determine whether the implemented mechanism produces the expected hydraulic, thermal or control response.

2 Physical Modelling and Preliminary Findings

2.1 Poor Balancing

Poor balancing concerns the distribution of the required water flow through the network. Danfoss describes hydraulic balance in district cooling in terms of supplying the required flow to the individual parts of the system under the relevant operating conditions [1]. The healthy condition is therefore not one fixed flow value because the flow required by an ETS changes with its cooling demand. In the present model, healthy operation is represented by the actual flow following the flow required for the current load,

$$\dot{V}_{\text{actual}}(t) \approx \dot{V}_{\text{required}}(t). \tag{1}$$

Poor balancing was represented by reducing the applied flow relative to this dynamically calculated requirement,

$$\dot{V}_{\text{applied}}(t) = m_b \dot{V}_{\text{required}}(t), \tag{2}$$

where m_b is the applied flow multiplier. For the initial mechanism test, $m_b = 0.80$ was used. This means that the affected ETS receives approximately 80% of its required flow. The value is a simulation sensitivity level rather than a proposed threshold for poor balancing in a real system.

Figure 1 shows the result. The required flow changes with the consumer load, while the applied flow follows the same overall pattern but remains below the requirement.

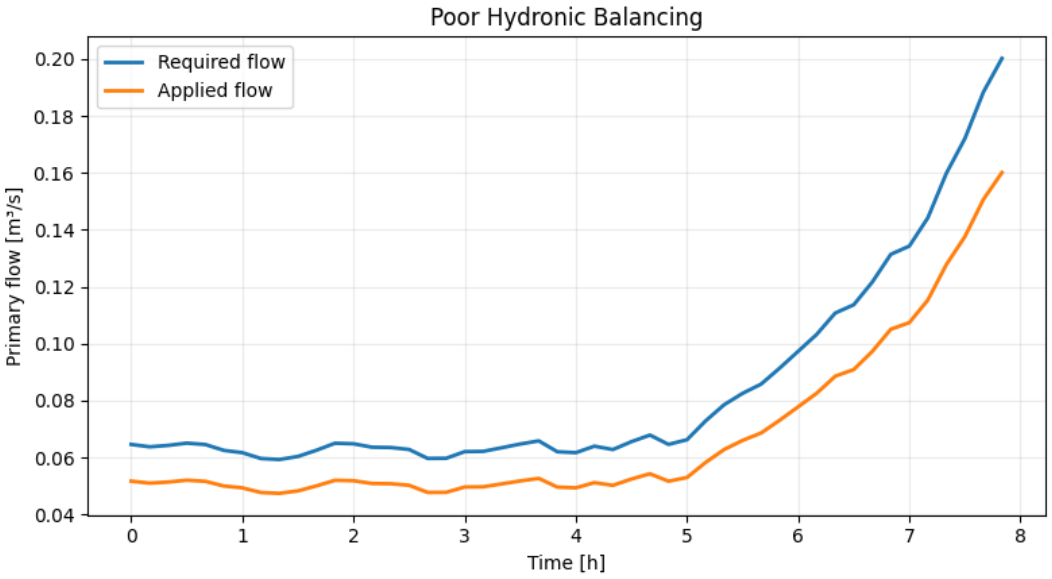


Figure 1: Required and applied primary flow during the poor balancing mechanism test. The applied flow was set to 80% of the dynamically calculated required flow.

To check whether the intervention behaved as intended, the normalised flow difference was calculated as

$$r_Q(t) = \frac{\dot{V}_{\text{applied}}(t) - \dot{V}_{\text{required}}(t)}{\dot{V}_{\text{required}}(t)}. \tag{3}$$

For a multiplier of 0.80, the expected value is -0.20. The simulation produced approximately -0.20004, with an absolute difference of approximately 4.23×10^{-5} from the intended value. Both the healthy and affected simulations converged for all 48 time steps. The result shows that the intervention produces the intended load-dependent flow deficit rather than simply imposing a fixed low flow.

2.2 Too Safe Safety Margins in Control

The second SWiM FSED concerns control settings that are more conservative than required by the actual operating condition. In a variable flow cooling network, one possible example is maintaining a plant head or differential pressure setpoint above what is needed by the connected consumers. Differential pressure reset strategies have been investigated for variable chilled water systems to adjust pressure requirements as terminal conditions change [2, 3].

The inherited simulation framework calculates a reference plant head during plant sizing. The FSED was represented by increasing this reference according to

$$H_{\text{fault}} = m_H H_0, \tag{4}$$

where H_0 is the healthy plant head and m_H is the applied multiplier. The initial test used $m_H = 1.20$. The healthy plant head was approximately 29.663 m and the modified head was 35.595 m, corresponding to an increase of approximately 5.933 m. Again, the 1.20 multiplier is an experimental sensitivity level rather than a proposed real-world threshold.

Figure 2 compares the healthy and increased plant head with the corresponding hydraulic pumping requirement. The consumer flow remained essentially unchanged because the ideal ETS flow controllers were still able to maintain the requested flow.

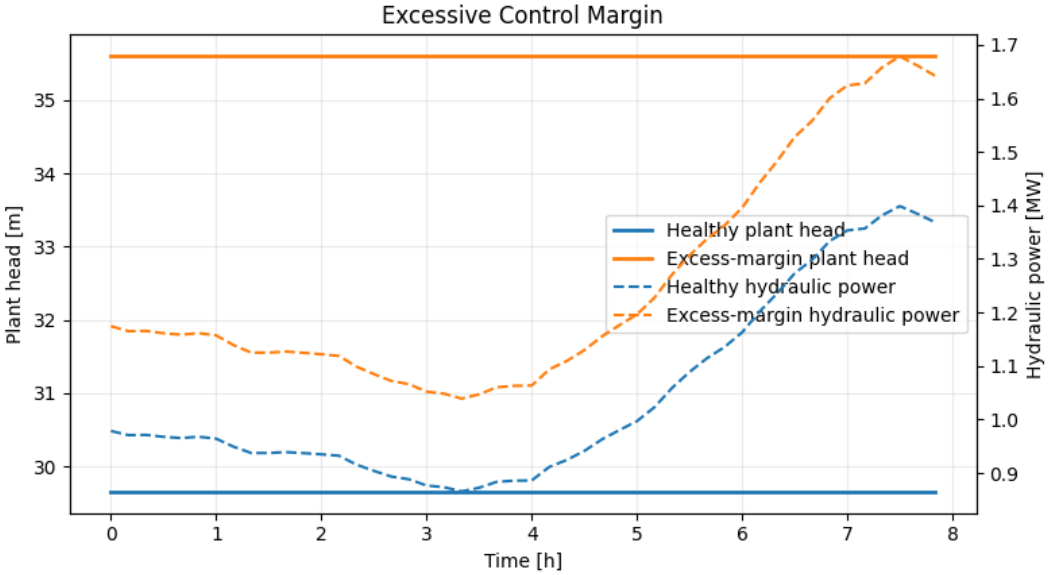


Figure 2: Plant head and calculated hydraulic pumping requirement for the healthy case and the test with a 20% increase in plant head.

Hydraulic pumping power was evaluated using

$$P_{\text{hyd}} = \rho g Q H, \tag{5}$$

where ρ is the water density, g is gravitational acceleration, Q is the volumetric flow and H is hydraulic head. The calculated hydraulic pumping requirement increased from approximately 1.041 MW to 1.249 MW, giving a ratio of approximately 1.20. This is consistent with the imposed 20% increase in head while plant flow remains almost unchanged. Both simulations converged for all 48 time steps.

The important observation is that the simulated system can continue delivering the requested consumer flow even when more hydraulic head is supplied than required. In this case, the FSED is therefore more visible in the relationship between supplied head and hydraulic requirement than in a loss of consumer flow.

2.3 Clogging of Water Filters

A filter or strainer introduces local hydraulic resistance into the water path. As material accumulates, the pressure difference required to maintain a comparable flow increases. Caleffi similarly uses differential pressure across a hydronic strainer as an indication of increasing restriction and the need for cleaning [4]. The inherited DCN model did not contain an explicit filter component for this FSED, so a local hydraulic resistance was introduced immediately upstream of the affected ETS flow control device.

The filter behaviour was represented using the local head loss relationship

$$h_L = K\frac{v^2}{2g}, \tag{6}$$

where h_L is local head loss, K is the local loss coefficient, v is water velocity and g is gravitational acceleration. The clean and clogged simulations used the same topology, geometry, load and operating conditions. Only the local loss coefficient was changed. The initial test used $K_{\text{clean}} = 1$ and $K_{\text{clogged}} = 20$. These values are simulation sensitivity levels and are not proposed as field thresholds for filter clogging.

Figure 3 shows the resulting flow and head loss. Mean filter flow remained approximately 0.08437 m^3/s in both simulations, while mean head loss increased from approximately 0.02048 m to 0.40939 m.

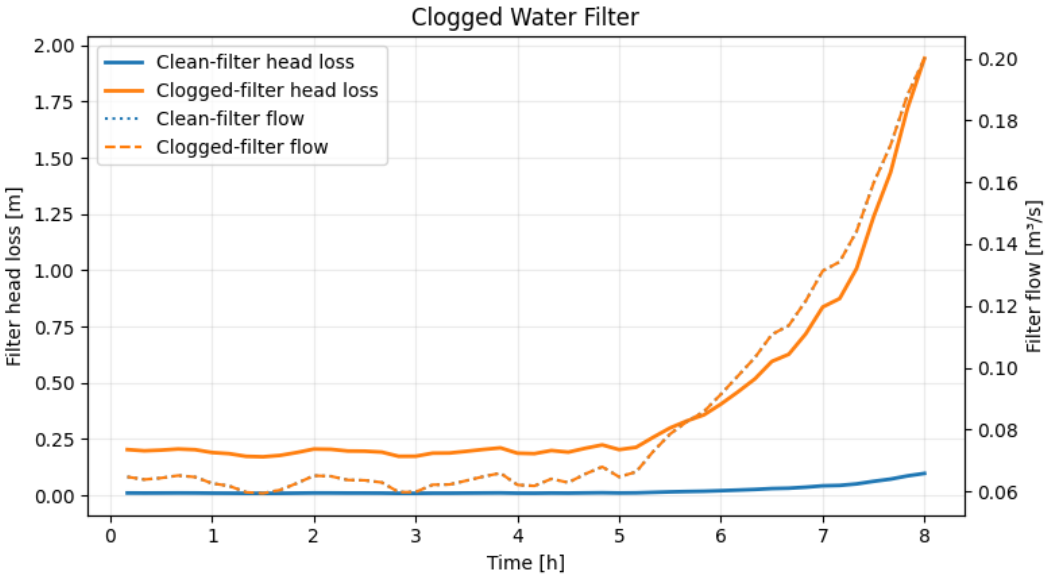


Figure 3: Filter flow and head loss for the clean and clogged water filter mechanism test. The local loss coefficient was increased from $K = 1$ to $K = 20$.

The observed head loss ratio was approximately

$$\frac{0.40939}{0.02048} \approx 19.986. \tag{7}$$

At comparable velocity, increasing K from 1 to 20 predicts an approximately twentyfold increase in local head loss from $h_L = Kv^2/(2g)$. The simulated ratio therefore closely follows the expected hydraulic relationship. The consumer flow remained almost unchanged because the ideal flow controller still had sufficient hydraulic authority to maintain the requested flow. Both simulations converged for all 48 time steps.

This result also shows why consumer flow alone is not necessarily the most useful signature for this mechanism in the current simulation. The stronger physical response is the increase in pressure loss across the filter while a similar flow is maintained.

2.4 Scaling/Fouling of Heat Exchangers

Scaling or fouling of a heat exchanger introduces additional thermal resistance at the heat transfer surfaces, reducing its ability to transfer heat. Fouling can also influence hydraulic resistance [5]. Experimental work by Genić et al. investigated fouling factors in plate heat exchangers used in district heating systems [6], while Guelpa and Verda investigated automatic fouling detection using operational measurements from district heating substations [7].

The thermal fouling representation was already present in the inherited `dcngen` framework [9]. It is therefore treated here as an inherited reference mechanism rather than as a new implementation. The model represents fouling by reducing the effective installed heat transfer conductance,

$$(UA)_{\text{fault}} = m_{UA}(UA)_{\text{healthy}}. \tag{8}$$

For the initial mechanism test, $m_{UA} = 0.85$ was used. Effective UA decreased from approximately 8.237 MW/K in the healthy simulation to 7.002 MW/K in the fouled simulation, giving the intended ratio of approximately 0.85.

Figure 4 compares the requested primary flow and primary side temperature difference. Mean requested flow increased from approximately 0.13565 m^3/s to 0.13871 m^3/s, while mean temperature difference decreased from approximately 7.476 K to 7.330 K.

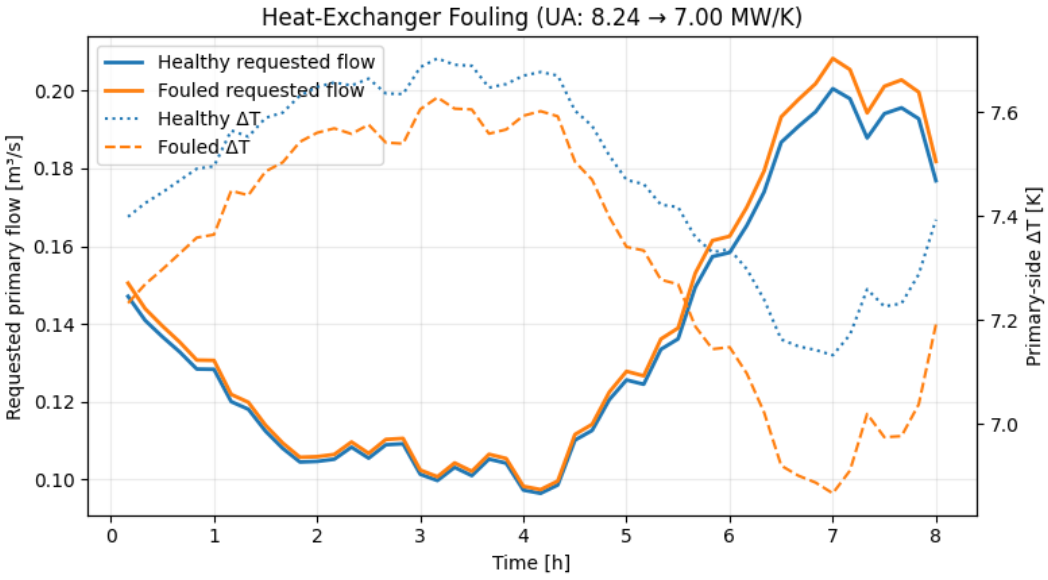


Figure 4: Requested primary flow and primary side temperature difference for the healthy and fouled heat exchanger simulations. Effective UA was reduced to 85% of the healthy value.

Both simulations converged for all 48 time steps. Of the 26 time steps that were feasible without unmet load in the matched healthy simulation, two became unmet after the fouling intervention. This distinction is important: the faulted simulation contained 24 unmet time steps in total, but 22 of these were already unmet in the corresponding healthy simulation and should not be attributed to fouling.

The observed response is consistent with reduced heat transfer capability. The affected ETS requests somewhat more primary flow and its thermal response changes after UA is reduced. Since this mechanism was inherited, it also provides a useful reference for checking whether the same validation approach can be applied to both inherited and newly implemented mechanisms.

2.5 Stuck/Sticking/Leaking Hydronic Valves

The fifth SWiM priority covers stuck, sticking and leaking hydronic valves. These behaviours are related but are not the same physical fault. The work completed so far addresses the *stuck* case. Valve stiction, which is more closely related to sticking behaviour, is associated with friction and stick-slip behaviour in a control valve. Choudhury et al. describe and model this phenomenon in control valves [8]. A leaking valve would require a different representation in which flow remains when the valve should be closed. Sticking and leaking are therefore not presented as completed implementations in this report.

The inherited simulator does not contain a detailed mechanical valve model with explicit stem position, actuator friction and valve coefficient C_v. Terminal flow is represented using an idealised flow control device. The implemented mechanism is therefore more precisely described as a stuck ETS flow control actuator. During healthy operation, the controller changes its requested primary flow as cooling demand changes. At fault onset t_f, the applied flow target is frozen at its last value,

$$\dot{V}_{\text{applied}}(t) = \dot{V}_{\text{applied}}(t_f), \qquad t \geq t_f, \tag{9}$$

while the controller continues calculating the flow required by the changing load.

Figure 5 shows the resulting behaviour. The mechanism test used consumer 10 and introduced the fault after four hours. Before the fault, the controller requested flow and applied flow follow one another. After the fault begins, the applied flow remains fixed while the requested flow continues changing.

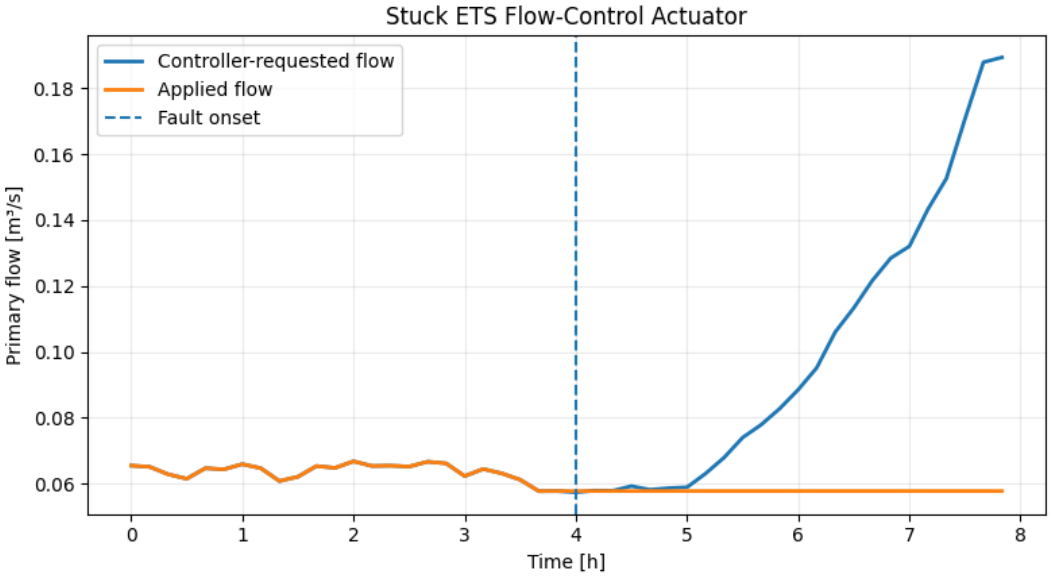


Figure 5: Controller requested and applied primary flow for the stuck case of the hydronic valve FSED. The dashed vertical line marks fault onset at four hours.

Before fault onset, the maximum difference between the matched healthy and faulted applied flow trajectories was approximately 2.78×10^{-17} m^3/s, which is effectively zero at simulation precision. After onset, the applied flow remained fixed while the desired flow continued changing. The post fault desired flow range was approximately 0.13197 m^3/s and the maximum difference between desired and applied flow reached approximately 0.13159 m^3/s. Both simulations converged for all 48 time steps.

The main signature of the implemented stuck case is therefore the disagreement between the

controller request and the applied flow. It should not simply be described as "low flow", because the requested flow itself changes with the consumer load.

3 Summary of Preliminary Findings

The mechanism tests produced distinct responses for the five FSED areas. Table 1 summarises what was changed in the simulator and the main response observed in each experiment. Four cases required implementation or extension of the inherited framework, while heat exchanger fouling was already available and was used as a reference mechanism.

Table 1: Summary of the mechanism tests completed so far. "Passed" refers to the simulation mechanism check described in this report and not to validation against field measurements.

SWiM FSED	**Simulation representation**	**Main preliminary finding**	**Check**
Poor balancing	Applied flow reduced relative to required flow	A load dependent deficit appeared between required and applied flow; the normalised difference closely matched the imposed multiplier.	Passed
Too safe safety margins in control	Plant head increased by 20%	Consumer flow was maintained while the calculated hydraulic pumping requirement increased by approximately 20%.	Passed
Clogging of water filters	Local loss coefficient increased from $K = 1$ to $K = 20$	Filter head loss increased by a factor of approximately 19.986 while comparable flow was maintained.	Passed
Scaling/fouling of heat exchangers	Effective UA reduced to 85%	Requested primary flow increased and primary side temperature difference decreased.	Passed
Stuck/sticking/leaking hydronic valves	Stuck case represented by freezing the ETS applied flow target	After fault onset, the applied flow remained fixed while the controller continued changing its requested flow.	Passed for stuck case

The comparison also suggests that there is no single variable that describes all five FSEDs. Poor balancing is most directly visible through required and delivered flow, too safe safety margins through required and supplied hydraulic head, filter clogging through pressure loss at comparable flow, heat exchanger fouling through thermal performance and flow response, and the stuck actuator through disagreement between controller request and applied flow. These mechanism specific relationships are useful when deciding which variables should later be retained in the synthetic dataset and fault detection experiments.

4 Current Synthetic Data Generation

Following the individual mechanism tests, work has started on combining the five mechanisms into a consistent synthetic data generation procedure. The planned pilot design uses three dynamic scenario seeds and three sensitivity settings for each of the five FSEDs, giving 45 healthy/fault scenario pairs. Each scenario contains 48 time steps. The same underlying scenario is used for the healthy and affected member of each pair so that the physical effect of the intervention can be checked before the data are used for detection.

The sensitivity settings currently being considered are flow multipliers of 0.90, 0.80 and 0.70 for poor balancing; plant head multipliers of 1.10, 1.20 and 1.30 for too safe safety margins in control; local filter loss coefficients of 5, 10 and 20 for clogging of water filters; and UA multipliers of 0.85, 0.70 and 0.50 for heat exchanger fouling. For the stuck case, the planned variation concerns

fault onset at different parts of the scenario rather than interpreting onset time as fault severity.

These values are used to explore the simulator response over different intervention strengths. They are not diagnostic limits derived from field measurements. The next dataset level check is to determine whether the expected physical response remains visible across these sensitivity settings and across different load realisations.

References

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